



Student Number:

Teacher:

St George Girls High School

Mathematics Extension 2

2023 Trial HSC Examination

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in **Section I**, use the Multiple-Choice answer sheet provided

For questions in **Section II**:

- Answer the questions in the booklets provided
- Start each question in a new writing booklet
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for incomplete or poorly presented solutions, or where multiple solutions are provided

Total marks:
100

Section I – 10 marks (pages 3 – 6)

- Attempt Questions 1– 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7 –12)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Q1-10	/10
Q11	/15
Q12	/16
Q13	/15
Q14	/15
Q15	/15
Q16	/14
TOTAL	/100
	%

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1 to 10.

1. What is the value of $\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right) e^{i\frac{\pi}{3}}$?

(A) $e^{i\frac{11\pi}{12}}$

(B) $\frac{1}{\sqrt{2}} e^{i\frac{-11\pi}{12}}$

(C) $e^{i\frac{\pi}{4}}$

(D) $e^{i\frac{-11\pi}{12}}$

2. Each pair of lines given below intersects at $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

Which pair of lines are perpendicular?

(A) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$

(B) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}$

(C) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

(D) $\ell_1: \underset{\sim}{r} = \begin{pmatrix} 3 \\ -2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \\ -5 \end{pmatrix}$ and $\ell_2: \underset{\sim}{r} = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$

3. Consider this statement.

“If I don’t make my bed then my mum will take my iPhone away from me.”

Which of the following is the converse of the contrapositive of the above statement?

- (A) “If I don’t make my bed then my mum will not take my iPhone away from me.”
- (B) “If my mum takes my iPhone away from me then I won’t make my bed”
- (C) “If I do make my bed then my mum will not take my iPhone away from me.”
- (D) “If I do make my bed then my mum will take my iPhone away from me.”

4. Vectors $\mathbf{u}$ and $\mathbf{v}$ have components $\begin{pmatrix} 0.6 \\ 0 \\ t \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 0 \\ -5 \end{pmatrix}$ respectively.

The following two statements are made about $\mathbf{u}$ and $\mathbf{v}$:

- (1) when $t = -1$, $\mathbf{u}$ and $\mathbf{v}$ are parallel.
- (2) when $t = -0.8$, $\mathbf{u}$ is a unit vector.

Which of the following is true?

- (A) Neither statement is correct.
- (B) Only statement (1) is correct.
- (C) Only statement (2) is correct.
- (D) Both statements are correct.

5. Which expression is equal to $\int \frac{1}{\sqrt{-16x - 4x^2}} dx$?

- (A) $\frac{1}{2} \sin^{-1} \left(\frac{x+2}{4} \right) + c$
- (B) $\frac{1}{4} \sin^{-1} \left(\frac{x-2}{2} \right) + c$
- (C) $\frac{1}{4} \sin^{-1} \left(\frac{x+2}{2} \right) + c$
- (D) $\frac{1}{2} \sin^{-1} \left(\frac{x+2}{2} \right) + c$

6. Evaluate $(1 + i)^{40} + (1 - i)^{40}$.

(A) 2^{21}

(B) 2^{20}

(C) 2^{19}

(D) 2^{18}

7. Using a suitable trigonometric substitution, $\int_0^{\frac{\pi}{3}} \cot^3 x \sec^2 x \, dx$ can be expressed in which of the following ways?

(A) $\int_0^{\sqrt{3}} u^3 \, du$

(B) $\int_0^{\frac{\pi}{3}} \frac{1}{u^3} \, du$

(C) $\int_0^{\frac{1}{\sqrt{3}}} u^{-3} \, du$

(D) $\int_0^{\sqrt{3}} \frac{1}{u^3} \, du$

8. Which of the following statements is correct?

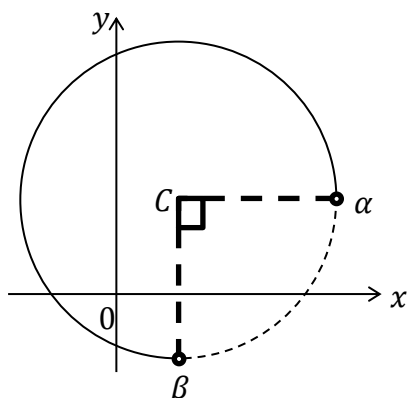
(A) $\forall a, b \in \mathbb{R} \quad \sin a < \sin b \Rightarrow a < b$

(B) $\forall a, b \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \sin a < \sin b \Rightarrow a < b$

(C) $\forall a, b \in \mathbb{R} \quad \cos a < \cos b \Rightarrow a < b$

(D) $\forall a, b \in [0, \pi] \quad \cos a < \cos b \Rightarrow a < b$

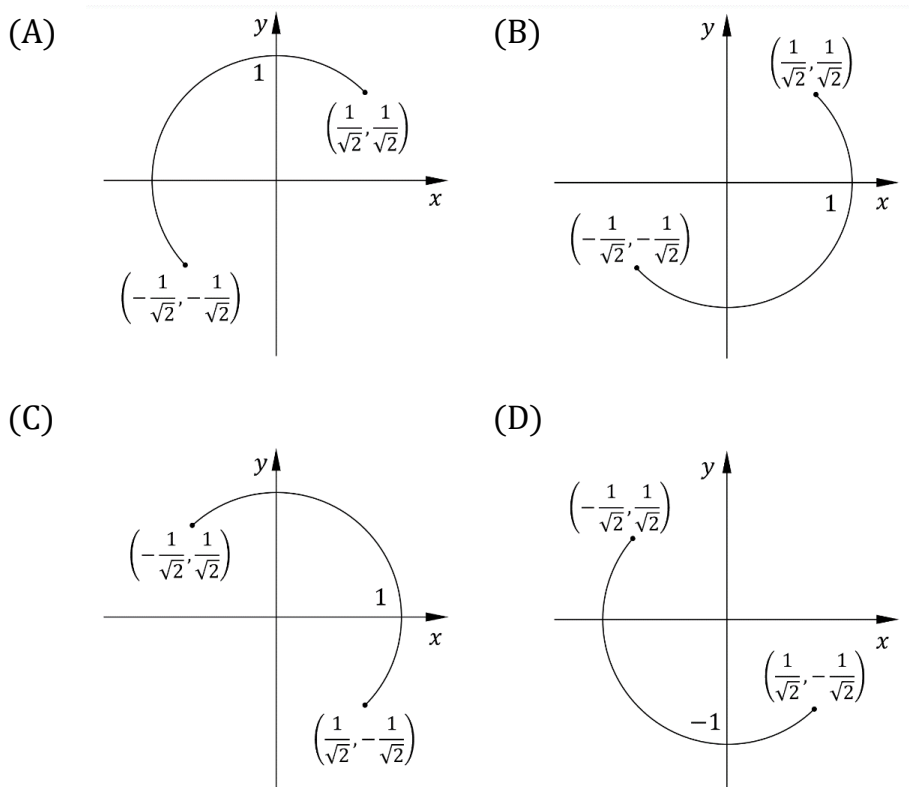
9. The diagram shows the solution of an equation, where C is the centre of the circle.



Which of these could be the equation?

- (A) $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = 0$
 (B) $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = \frac{\pi}{2}$
 (C) $\text{Arg}(z - \alpha) - \text{Arg}(z - \beta) = \frac{\pi}{4}$
 (D) $\text{Arg}(z - \beta) - \text{Arg}(z - \alpha) = \frac{\pi}{4}$
10. Which diagram best shows the curve described by the position vector

$$\tilde{r}(t) = \sin(t) \tilde{i} - \cos(t) \tilde{j} \text{ for } \frac{\pi}{4} \leq t \leq \frac{5\pi}{4}?$$



Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations

Question 11 (15 marks) Use a SEPARATE writing booklet. **Marks**

(a) Consider the complex numbers $z_1 = 5 + i$ and $z_2 = -2 + i$.

Find the value of $\frac{\overline{z_1}}{z_1 + z_2}$, giving your answer in the form $a + ib$. 2

(b) For the complex number $z = 3e^{i\frac{\pi}{6}}$, write $\overline{z}^4$ in the form $a + ib$. 2

(c) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{3x}{\sqrt{x^2 + 2}} dx$. 3

(d) (i) Find the two square roots of $2i$, giving your answers in the form $x + iy$, where x and y are real numbers. 2

(ii) Hence, solve $2z^2 + 2\sqrt{2}z + 1 - i = 0$. Give your answers in the form $x + iy$. 2

(e) Consider the two vectors $\underline{u} = 2\alpha \underline{i} + (3 - \alpha)\underline{k}$ and $\underline{v} = 2\underline{i} - 2\underline{j} + \underline{k}$, where α is a scalar.

(i) Find p , the vector projection of $\underline{u}$ onto $\underline{v}$. 2

(ii) Given that $3|\underline{u}| = \sqrt{17}|\underline{p}|$, find the value of α . 2

Question 12 (16 marks) Use a SEPARATE writing booklet.	Marks
(a) Use mathematical induction to prove that for all positive odd integers n , $n(n^2 + 1)$ is even.	3
(b) The polynomial $g(z) = z^4 + 2z^3 + 6z^2 + 8z + 8$ has roots $a + bi$ and $2ci$, where a, b and c are all real.	
(i) Find all the roots of $g(z)$.	3
(ii) Write $g(z)$ as a product of two real quadratic factors.	2
(c) (i) Find the values of a, b , and c such that:	3
$\frac{2x}{(x-4)(x+2)^2} = \frac{a}{x-4} + \frac{b}{x+2} + \frac{c}{(x+2)^2}.$	
(ii) Hence find $\int \frac{x}{(x-4)(x+2)^2} dx$.	2
(d) Shade the region on the Argand diagram where the inequalities $ z - 2 - 2i \leq 2$ and $ \operatorname{Im}(z - 2i) \geq 1$ hold simultaneously.	3

Question 13 (15 marks) Use a SEPARATE writing booklet.	Marks
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- | | |
|---|---|
| (a) (i) Find all solutions to the equation $z^5 = -1$. | 3 |
|---|---|

Give your answers in modulus-argument form.

- | | |
|--|---|
| (ii) Hence prove that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$. | 2 |
|--|---|

- | | |
|--|---|
| (b) Use integration by parts to find $\int x e^{-2x} dx$. | 3 |
|--|---|

- | | |
|---|---|
| (c) Prove by contradiction that if p is an integer, then $p^2 + 6$ is not divisible by 4. | 3 |
|---|---|

- | | |
|---|--|
| (d) The line l has equation $\tilde{v} = \begin{pmatrix} 0 \\ 3 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ b \\ 2 \end{pmatrix}$, where μ is a parameter and b is a constant. | |
|---|--|

The points P and Q have coordinates $(-1, 2, 3)$ and $(-2, 4, 2)$ respectively.

- | | |
|--|---|
| (i) Find a vector equation of the line PQ.
[Leave your answer in the form $\tilde{r} = a + \lambda b$] | 1 |
|--|---|

- | | |
|--|---|
| (ii) Find the value of b for which the acute angle between line l and the line PQ is $\cos^{-1} \frac{1}{6}$. | 3 |
|--|---|

Question 14 (15 marks) Use a SEPARATE writing booklet.	Marks
(a) Calculate the modulus and argument of the sum of the roots of the equation $(4 + 3i)z^2 - (3 - i)z - (4 + 2i) = 0$ in exact form.	3
(b) (i) If $t = \tan \frac{\theta}{2}$, show that $\frac{d\theta}{dt} = \frac{2}{1+t^2}$.	2
(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{3}{8 \cos \theta + 10} d\theta$. Leave your answer correct to 3 significant figures.	3
(c) Consider the line L with position vector $\underset{\sim}{r} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}.$ The sphere with equation $(x - 3)^2 + (y + 2)^2 + (z - 4)^2 = 18$ intersects the line L at the points A and B. Find the equation of the sphere with diameter AB.	4
(d) Use mathematical induction to prove that $n! \geq 2^{n-1}$, $n \in \mathbb{Z}^+$.	3

Question 15 (15 marks) Use a SEPARATE writing booklet.

Marks

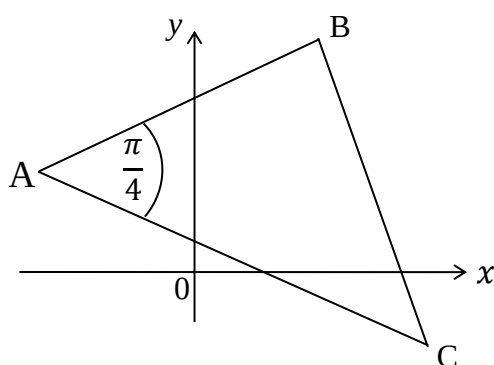
- (a) Prove by induction that $T_n = 3(2^n) + 1$ for $n \geq 1$, given $T_1 = 7$ and $T_n = 2T_{n-1} - 1$ for $n \geq 2$.

3

- (b) The vertices A, B and C of triangle ABC are represented in the Argand diagram by the complex numbers a , b and c respectively.

3

$$AC = \sqrt{2} AB \text{ and } \angle CAB = \frac{\pi}{4}.$$



By using vectors, or otherwise, show that $c = b(1 - i) + ai$.

- (c) Using the substitution $x = 2 + 2 \cos^2 \theta$, calculate the value of

4

$$\int_2^3 \sqrt{\frac{x-2}{4-x}} dx.$$

- (d) (i) Let $I_n = \int_0^{\frac{\pi}{2}} x^n \cos x dx$ for $n = 0, 1, 2, 3, \dots$

Show that $I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$ for every integer $n \geq 2$.

3

- (ii) Hence, show that $I_4 = \left(\frac{\pi}{2}\right)^4 - 12\left(\frac{\pi}{2}\right)^2 + 24$.

2

Question 16 (14 marks) Use a SEPARATE writing booklet.

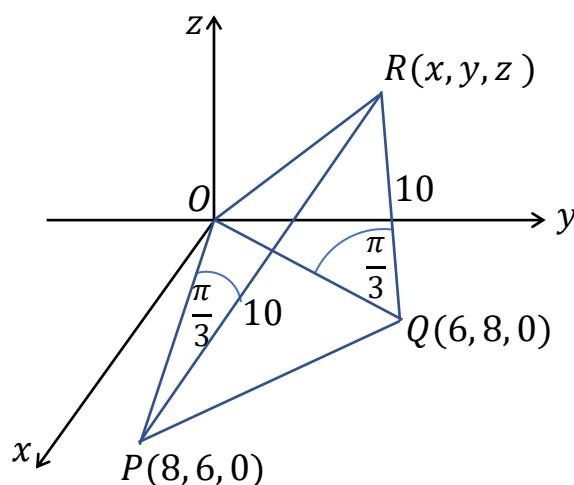
Marks

- (a) (i) Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$. 2

- (ii) By choosing a suitable trigonometric substitution, determine the value of 3

$$\int_0^1 \frac{dx}{x + \sqrt{1-x^2}}.$$

- (b) The diagram shows a triangular pyramid with vertices $O(0, 0, 0)$, $P(8, 6, 0)$, $Q(6, 8, 0)$ and $R(x, y, z)$, where x, y and z are positive real numbers.



- Given that $\angle RPO = \angle RQO = \frac{\pi}{3}$, $|\overrightarrow{PR}| = |\overrightarrow{QR}| = 10$ units, find the coordinates of R . 4

- (c) (i) Using the inequality $\frac{x+y}{2} \geq \sqrt{xy}$, show that 3

$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 \geq 3(\sqrt{ab} + \sqrt{ac} + \sqrt{bc}),$$

where a, b and c are positive numbers.

- (ii) Hence, or otherwise, show that 2

$$(m^3p^3 + m^3r^3 + p^3r^3)^2 \geq m^3p^3r^3(m^3 + r^3 + p^3).$$

END OF EXAMINATION

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SGGHS 2023 Mathematics Extension 2 Trial HSC Examination

SUGGESTED SOLUTIONS

Section 1

Multiple Choice Answer Key

Question	Answer
1	D
2	A
3	C
4	D
5	D
6	A
7	D
8	B
9	C
10	C

Section I

$$1. e^{\frac{3\pi}{4}i} \times e^{\frac{\pi}{2}i} = e^{(\frac{3\pi}{4} + \frac{\pi}{2})i} \\ = e^{\frac{5\pi}{4}i} \\ = e^{-\frac{3\pi}{4}i}$$

D

2. Looking for the dot product of the direction vectors being zero.

$$A: \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \\ = -2 + 2 + 0 \\ = 0 \quad \text{so A}$$

A

3. The contrapositive of the statement is:

'If my mum does not take the iPhone away from me, then I will make my bed.'

Hence, the **converse** of the contrapositive is:

'If I do make my bed, then my mum will not take the iPhone away from me.'

C

$$4. \quad \vec{u} = \begin{pmatrix} 0.6 \\ 0 \\ t \end{pmatrix} \quad \vec{v} = \begin{pmatrix} 3 \\ 0 \\ -5 \end{pmatrix} \\ \text{when } t = -1 \quad \vec{u} = \begin{pmatrix} 0.6 \\ 0 \\ -1 \end{pmatrix} \\ = \begin{pmatrix} \frac{3}{5} \\ 0 \\ -1 \end{pmatrix} \\ \text{Now } 5 \times \begin{pmatrix} \frac{3}{5} \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -5 \end{pmatrix} \quad \therefore (1) \text{ true} \\ \text{when } t = -0.8, \quad \vec{u} = \begin{pmatrix} 0.6 \\ 0 \\ -0.8 \end{pmatrix}$$

$$|\vec{u}| = \sqrt{(0.6)^2 + (0.8)^2} = 1 \quad \therefore \text{unit vector } \therefore (2) \text{ true}$$

$\therefore$ Both statements are true.

D

$$5. \quad \int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c \\ \int \frac{1}{\sqrt{-16x - 4x^2}} dx = \int \frac{1}{2\sqrt{-4x - x^2}} dx \\ = \int \frac{1}{2\sqrt{4 - 4x - x^2}} dx \\ = \int \frac{1}{2\sqrt{2^2 - (4 + 4x + x^2)}} dx \\ = \int \frac{1}{2\sqrt{2^2 - (x+2)^2}} dx \\ = \frac{1}{2} \sin^{-1} \left(\frac{x+2}{2} \right) + c$$

D

$$6. \quad z = 1 + i, \quad \bar{z} = \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$\bar{z} = 1 - i = \sqrt{2} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right).$$

Using de Moivre's theorem $z^{40} = 2^{20} \text{cis}(10\pi)$,

$$(\bar{z})^{40} = 2^{20} \text{cis}(-10\pi).$$

$$z^{40} + (\bar{z})^{40} = z^{40} + \overline{z^{40}}$$

$$= 2 \text{Re}(z^{40}) = 2^{21} \cos(10\pi).$$

$$\therefore (1+i)^{40} + (1-i)^{40} = 2^{21} \cos(10\pi). \\ = 2^{21}$$

A

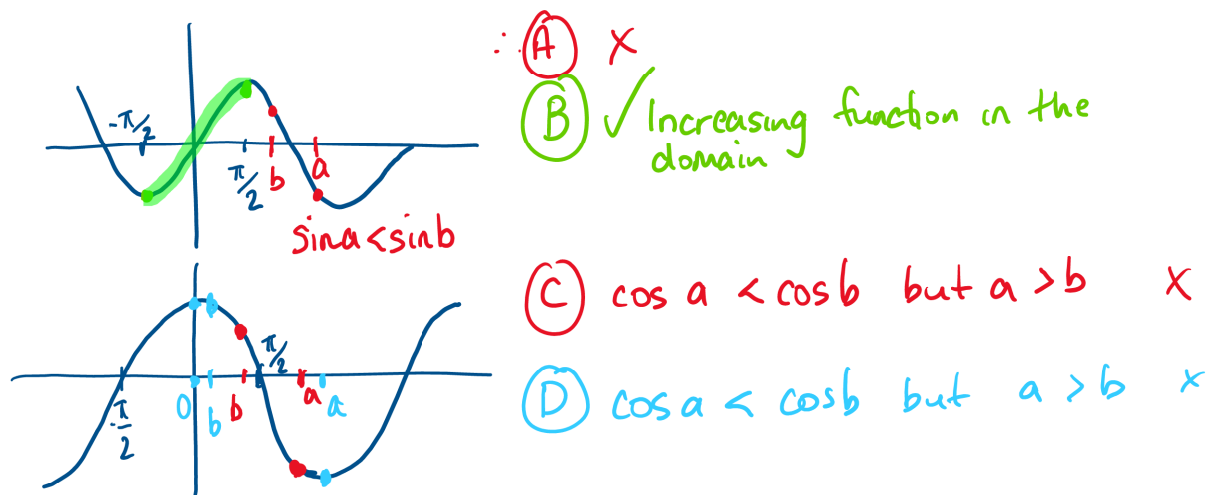
$$\begin{aligned}
 7. \quad & \int_0^{\frac{\pi}{3}} \cot^3 x \sec^2 x \, dx \\
 &= \int_0^{\frac{\pi}{3}} \frac{1}{\tan^3 x} \sec^2 x \, dx \\
 &= \int_0^{\sqrt{3}} \frac{1}{u^3} \, du
 \end{aligned}$$

Let $u = \tan x$
 $du = \sec^2 x$
 when $x = 0$ $u = \tan 0 = 0$
 when $x = \frac{\pi}{3}$ $u = \tan \frac{\pi}{3} = \sqrt{3}$

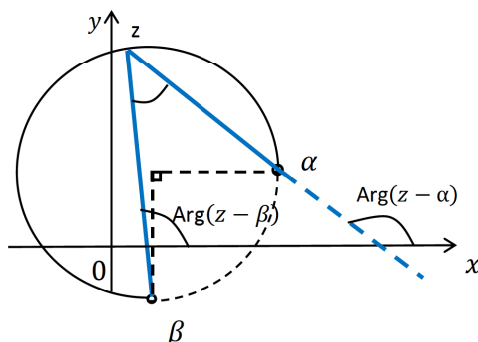
D

8. The correct answer must be an increasing function in the whole domain, which only occurs for B.

B



9.



z lies on the circumference of a circle, radii from α and β meet at the centre at $\frac{\pi}{2}$ radians, this means the angle at z must be $\frac{\pi}{4}$ radians as the angle at circumference is half the angle at the centre when subtended by the same arc.

$z - \alpha$ represents the vector from α to z .

$z - \beta$ represents the vector from β to z .

Using the exterior angle of a triangle, we can see that the angle at $z = \text{Arg}(z - \alpha) - \text{Arg}(z - \beta)$.

C

$$\begin{aligned}
 10. \quad & \tilde{r}\left(\frac{\pi}{4}\right) = \sin \frac{\pi}{4} \times \tilde{i} - \cos \frac{\pi}{4} \times \tilde{j} = \frac{1}{\sqrt{2}} \tilde{i} - \frac{1}{\sqrt{2}} \tilde{j} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) \\
 & \tilde{r}\left(\frac{5\pi}{4}\right) = \sin \frac{5\pi}{4} \times \tilde{i} - \cos \frac{5\pi}{4} \times \tilde{j} = -\frac{1}{\sqrt{2}} \tilde{i} - \left(-\frac{1}{\sqrt{2}}\right) \tilde{j} = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)
 \end{aligned}$$

Also passes through:

$$\tilde{r}\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} \times \tilde{i} - \cos \frac{\pi}{2} \times \tilde{j} = \tilde{i} = (1, 0), \text{ so C.}$$

C

MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $\bar{z} = 5 - i$		This part was very well done
$\frac{\bar{z}}{z_1 + z_2} = \frac{5 - i}{5 + i + -2 + i}$		
$= \frac{5 - i}{3 + 2i}$		
$= \frac{5 - i}{3 + 2i} \times \frac{3 - 2i}{3 - 2i}$	1	
$= \frac{15 - 10i - 3i - 2}{9 + 4}$		
$= \frac{13 - 13i}{13}$		
$= \frac{13(1 - i)}{13}$		
$= 1 - i$	1	(2)
b) $z = 3e^{\frac{\pi}{6}i}$		
$= 3(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$		
$\bar{z} = 3(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})$		
$\bar{z}^4 = [3(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})]^4$		
$= 3^4 (\cos \frac{4\pi}{6} - i \sin \frac{4\pi}{6})$	1	
$= 81 (\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3})$	1/2	
$= 81 (-\frac{1}{2} - i \frac{\sqrt{3}}{2})$	1/2	
$= -\frac{81}{2} - \frac{81\sqrt{3}i}{2}$		(2)

MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) Let $I = \int_{\sqrt{2}}^{\sqrt{3}} \frac{3x}{\sqrt{x^2+2}} dx$ $= \frac{3}{2} \int_{\sqrt{2}}^{\sqrt{3}} \frac{2x}{\sqrt{x^2+2}} dx$ Let $u = x^2$ $\frac{du}{dx} = 2x$ $du = 2x dx$ when $x = \sqrt{3}$, $u = 3$ $x = \sqrt{2}$, $u = 2$ $\therefore I = \frac{3}{2} \int_2^3 \frac{du}{\sqrt{u+2}} \quad \frac{1}{2}$ $= \frac{3}{2} \int_2^3 (u+2)^{-\frac{1}{2}} du$ $= \frac{3}{2} \left[2(u+2)^{\frac{1}{2}} \right]_2^3 \quad \frac{1}{2}$ $= 3(\sqrt{3+2} - \sqrt{2+2})$ $= 3(\sqrt{5} - \sqrt{4})$ $= 3\sqrt{5} - 3 \times 2$ $= 3\sqrt{5} - 6$ Alternative method. $I = \frac{3}{2} \int_{\sqrt{2}}^{\sqrt{3}} \frac{2x}{\sqrt{x^2+2}} dx$ $= \frac{3}{2} \int_{\sqrt{2}}^{\sqrt{3}} 2x(x^2+2)^{-\frac{1}{2}} dx \quad \frac{1}{2}$ $= \frac{3}{2} \left[\frac{(x^2+2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_{\sqrt{2}}^{\sqrt{3}} \quad \frac{1}{2}$ $= \frac{3}{2} \left[2\sqrt{x^2+2} \right]_{\sqrt{2}}^{\sqrt{3}}$ $= 3 \left[\sqrt{(\sqrt{3})^2+2} - \sqrt{(\sqrt{2})^2+2} \right]$ $= 3(\sqrt{5} - \sqrt{4})$		Overall this question was done well. 3

MATHEMATICS EXTENSION 2 – QUESTION 11

[illegible]

MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
e) i) $\vec{u} = 2\alpha\vec{i} + (3-\alpha)\vec{k}$ $\vec{v} = 2\vec{i} - 2\vec{j} + \vec{k}$		
Now $p = \text{proj}_{\vec{v}} \vec{u}$		
$= \frac{\vec{u} \cdot \vec{v}}{ \vec{v} ^2} \times \vec{v}$	1/2 mk	for correct $\vec{u} \cdot \vec{v}$
$= \frac{2\alpha \times 2 + (3-\alpha) \times 1}{(\sqrt{2^2 + 2^2 + 1^2})^2} \times \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$		
$= \frac{4\alpha + 3 - \alpha}{4 + 4 + 1} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$	1/2 mk	for finding $ \vec{v} ^2$ Many students did not square $ \vec{v} $
$= \frac{3\alpha + 3}{9} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$		
$= \frac{3(\alpha + 1)}{9} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$		
$= \frac{\alpha + 1}{3} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$	1 mk	(2)
ii) $\vec{u} = \sqrt{(2\alpha)^2 + (3-\alpha)^2}$ $= \sqrt{4\alpha^2 + 9 - 6\alpha + \alpha^2}$ $= \sqrt{5\alpha^2 - 6\alpha + 9}$		
and $\vec{p} = \frac{\alpha + 1}{3} \times \sqrt{2^2 + (-2)^2 + 1}$	1/2 mk	Many students did not multiply $\sqrt{2^2 + (-2)^2 + 1}$ by $\frac{\alpha + 1}{3}$ and therefore have made the question easier.
$= \frac{\alpha + 1}{3} \times \sqrt{4 + 4 + 1}$		
$= \frac{\alpha + 1}{3} \times \sqrt{9}$		
$= \frac{\alpha + 1}{3} \times 3\sqrt{17} \quad \vec{p} $		
$= \alpha + 1 \quad (\alpha + 1)$		
Now $3 \vec{u} =$		
$3\sqrt{5\alpha^2 - 6\alpha + 9} = \sqrt{17}$		

MATHEMATICS EXTENSION 2 – QUESTION 11

[illegible]

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
<u>Step 1</u> - Prove true for $n=1$		
$x(x^2+1) \quad x \rightarrow n \quad n(n^2+1)$		
$n=1$		
$= 1(1^2+1)$		
$= 2$ which is even.	(1/2)	Correctly proves the base case
<u>Step 2</u> - Assume true for $n=k$ where k is odd		
i.e. Assume $k(k^2+1) = 2M$ where $M \in \mathbb{Z}$	(1/2)	Correctly states the assumption
<u>Step 3</u> - Prove true for $n=k+2$		
i.e. Prove $(k+2)[(k+2)^2+1]$ is even $k \in \text{odd integer}$		
$(k+2)[(k+2)^2+1]$	(1/2)	Correctly states what is being proved.
$= (k+2)(k^2+4k+4+1)$		
$= (k+2)(k^2+4k+5)$		
$= k^3+4k^2+5k+2k^2+8k+10$		
$= k^3+6k^2+13k+10$		
$= k^3+k+6k^2+12k+10$		
$= k(k^2+1) + 2(3k^2+6k+5)$		
$= 2M + 2(3k^2+6k+5)$ by the assumption	(1)	Correctly uses the assumption in the proof
$= 2(M+3k^2+6k+5)$ $M \in \mathbb{Z}$, k is a positive odd integer		
which is even	(1/2)	Correctly completes proof and includes conclusion-
$\therefore$ true for $n=k+2$, if true for $n=k$		
<u>Step 4</u>		
$\therefore$ By the Principal of Mathematical Induction		
it is true for all positive odd integers n .		

MATHEMATICS EXTENSION 2 – QUESTION		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
OK <u>step 3</u> Prove true for $n = 2k + 1$ i Prove $(2k+1)[(2k+1)^2 + 1]$ is even $(2k+1)[(2k+1)^2 + 1]$ $= (2k+1)(4k^2 + 4k + 1 + 1)$ $= 8k^3 + 8k^2 + 4k + 4k^2 + 4k + 2$ $= 8k^3 + 12k^2 + 8k + 2$ $= 8(2M - k) + 12k^2 + 8k + 2$ $= 16M - 8k + 12k^2 + 8k + 2$ $= 16M + 12k^2 + 2$ $= 2(8M + 6k^2 + 1)$ etc ... which is even $M \in \mathbb{Z}$, k is an odd positive integer		

Step 3 Prove true for $n = 2k + 1$

$$(2k+1)[(2k+1)^2+1]$$

$$= (2k+1)(4k^2 + 4k + 1 + 1)$$

$$= 8k^3 + 8k^2 + 4k + 4k^2 + 4k + 2$$

$$= 8k^3 + 12k^2 + 8k + 2$$

$$= 8(2M - k) + 12k^2 + 8k + 2$$

$$= 16M - 8k + 12k^2 + 8k + 2$$

$$= 16M + 12k^2 + 2$$

$$= 2(8M + 6k^2 + 1) \quad \text{which is even MEZ, } k \text{ is an odd positive integer}$$

etc...

$$k(k^2+1) = 2M$$

$$k^3 + k = 2M$$

$$k^3 = 2M - k$$

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b)(i) $g(z) = z^4 + 2z^3 + 6z^2 + 8z + 8$ $a, b, c \in \mathbb{R}$		
If $a+bi$ is a root then $a-bi$ is a root.		
If $2ci$ is a root then $-2ci$ is a root.		
This is because the roots occur in conjugate pairs when the coefficients are real.		
$\alpha + \beta + \gamma + \delta = -\frac{b}{a}$		
$a+ib + a-ib + 2ci - 2ci = -2$		to get all conjugate pairs and correctly evaluates
$2a = -2$		
$a = -1$	①	$a = -1$
$\alpha\beta\gamma\delta = \frac{c}{a}$		
$(a+ib)(a-ib)(2ci)(-2ci) = 8$	②	correct expression for product of roots
$4c^2(a^2+b^2) = 8 \quad \dots \quad \textcircled{1}$		
$\alpha\beta\gamma + \alpha\beta\delta + \beta\gamma\delta + \alpha\gamma\delta$	②	correct expression for either sum of the roots two at a time or three at a time.
$(a+ib)(a-ib)(2ci) + (a+ib)(a-ib)(-2ci) + (a-ib)(2ci)(-2ci) + (a+ib)(2ci)(-2ci) = -8$		
$2ci(a^2+b^2) - 2ci(a^2+b^2) + 4c^2(a-ib) + 4c^2(a+ib) = -8$		
$4ac^2 - 4bc^2i + 4ac^2 + 4bc^2i = -8$		
$8ac^2 = -8$		
$ac^2 = -1$		
$a = -1 \quad \therefore (-1)c^2 = -1$		
$c^2 = 1$		
$c = \pm 1$	②	Correctly evaluates $c = \pm 1$
$\therefore$ From $\textcircled{1} \quad 4c^2(a^2+b^2) = 8$		
$4c^2((-1)^2+b^2) = 8$		
$c^2(1+b^2) = 2$		
For $c = \pm 1 \quad (\pm 1)^2(1+b^2) = 2$		
$1+b^2 = 2$		
$b^2 = 1$		
$b = \pm 1$	②	Correctly evaluates $b = \pm 1$
Now roots are $a \pm bi, \pm 2ci$		
$-1 \pm i, \pm 2i \quad (\text{i.e. } -1+i, -1-i, 2i, -2i)$		* $\frac{1}{2}$ mark lost if roots are not written

MATHEMATICS EXTENSION 2 – QUESTION

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$\alpha \beta + \alpha \gamma + \alpha \delta + \beta \gamma + \beta \delta + \gamma \delta = 6$ $(a+ib)(a-ib) + (a+ib)(-2ci) + (2ci)(a+ib) + (a-ib)(2ci) + (a-ib)(-2ci) + (2ci)(-2ci) = 6$ $a^2 + b^2 - 2aci + 2bc + 2aci - 2bc + 2ci^2 + 2bc - 2aci - 2bc + 4c^2 = 6$ $a^2 + b^2 + 4c^2 = 6$ $(-1)^2 + b^2 + 4(\pm 1)^2 = 6$ $1 + b^2 + 4 = 6$ $b^2 = 1$ Note if you got $b = \pm\sqrt{3}$, $c = \pm\sqrt{2}$ $b = \pm 1$ These solutions do not give zeros when substituted in.		
(ii) $g(z) = [z - (-1+i)][z - (-1-i)](z-2i)(z+2i)$ $= (z+1-i)(z+1+i)(z-2i)(z+2i)$ $= [(z+1)-i][(z+1)+i](z^2+4)$ $= [(z+1)^2+1](z^2+4)$ $= (z^2+2z+2)(z^2+4)$ NOTE: $a^2 + b^2 + 4c^2 = 6$ $1 + 3 + 4c^2 = 6$ $4c^2 = 2$ $c^2 = \frac{1}{2}$ $c = \pm\frac{1}{\sqrt{2}}$ $P(\sqrt{2}i) = (\sqrt{2}i)^4 + 2(\sqrt{2}i)^3 + 6(\sqrt{2}i)^2 + 8(\sqrt{2}i) + 8$ $= 4 - 4\sqrt{2}i - 12 + 8\sqrt{2}i + 8 + 8\sqrt{2}i \neq 0$ $\neq 0$ $P(2i) = (2i)^4 + 2(2i)^3 + 6(2i)^2 + 8(2i) + 8$ $= 16 - 16i - 24 + 16i + 8$ $= 0$		

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) (i) $\frac{2x}{(x-4)(x+2)^2} = \frac{a}{x-4} + \frac{b}{x+2} + \frac{c}{(x+2)^2}$		
$2x = a(x+2)^2 + b(x-4)(x+2) + c(x-4)$	① mark	correct equation
Sub in $x = -2$		
$-4 = -6c$		
$c = \frac{2}{3}$	① mark	to find one of the values
Sub in $x = 4$		
$8 = 36a$		
$a = \frac{2}{9}$		
Sub in $x = 0$		
$0 = 4a - 8b - 4c$		
$0 = 4(\frac{2}{9}) - 8b - 4(\frac{2}{3})$		
$0 = 8 - 72b - 24$		
$72b = -16$		
$b = -\frac{2}{9}$		
$\therefore a = \frac{2}{9}, b = -\frac{2}{9}, c = \frac{2}{3}$	① mark	find all correct values
(ii) $\int \frac{x}{(x-4)(x+2)^2} dx = \frac{1}{2} \int \frac{2x}{(x-4)(x+2)^2} dx$	① mark	to establish correct integral to enable use of (i)
$= \frac{1}{2} \int \frac{2}{9(x-4)} dx - \frac{1}{2} \int \frac{2}{9(x+2)} dx + \frac{1}{2} \int \frac{2}{3(x+2)^2} dx$		
$= \frac{1}{9} \int \frac{1}{x-4} dx - \frac{1}{9} \int \frac{1}{x+2} dx + \frac{1}{3} \int (x+2)^{-2} dx$		
$= \frac{1}{9} \ln x-4 - \frac{1}{9} \ln x+2 + \frac{1}{3} \left(\frac{x+2}{-1} \right)^{-1} + c$		
$= \frac{1}{9} \ln \left \frac{x-4}{x+2} \right - \frac{1}{3(x+2)} + c$	①	Correct answer
		NOTE: Some half marks awarded for a minor error within working.

[illegible][illegible]

$\frac{dx}{dt} = \frac{a}{b} + \frac{c}{d}$		
---	--	--

$(x-4)(x+2)^2$	$x-4$	$x+2$	$(x+2)^2$		
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$$2x = a(x+2)^2 + b(x-4)(x+2) + c(x-4)$$

$$2x = ax^2 + 4ax + 4a + bx^2 - 2bx - 8b + cx - 4c$$

$$0x^2 + 2x + 0 = (a+b)x^2 + (4a-2b+c)x + 4a-8b-4c$$

By equating coefficients		
--------------------------	--	--

$a+b=0$	$4a-2b+c=2$	$4a-8b-4c=0$		
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$b = -a$	$4a + 2a + c = 2$	$a - 2b - c = 0$		
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$$6a + c = 2 \quad \leftarrow \quad a + 2a - c = 0$$

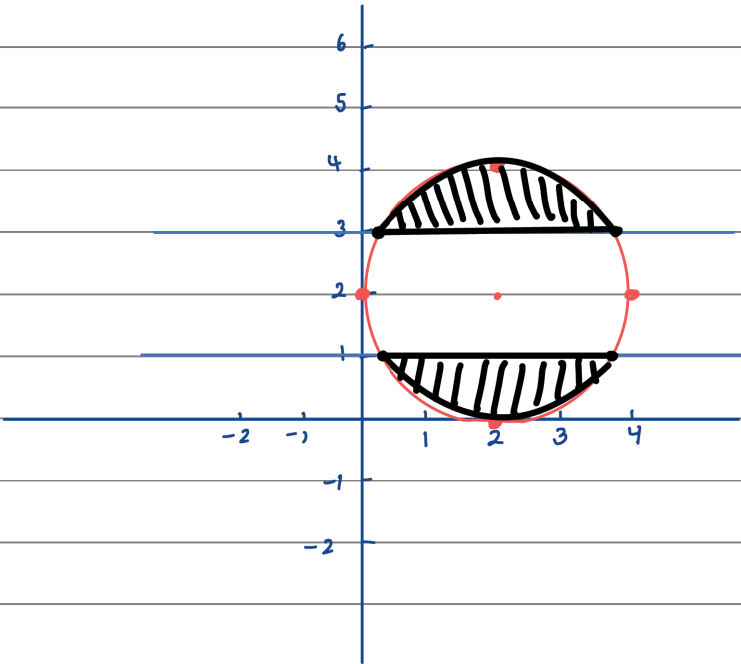
$$\underbrace{\quad}_{\text{sub}} \quad 3a = c$$

$$\therefore 6a + 3a = 2$$

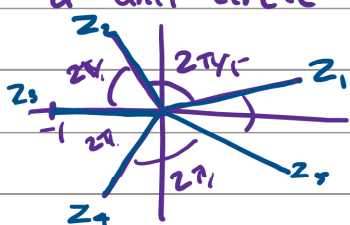
$9a=2$		

$$\therefore a = \frac{2}{9}, b = -\frac{2}{9}, c = \frac{2}{3}$$

MATHEMATICS EXTENSION 2 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) $z - (2+2i) \leq 2$ Circle centre $(2,2)$, radius=2 		
	① mark	correct circle
	① mark	one correct line + shading
	① mark	Correct graph
$\operatorname{Im}(z - 2i) \geq 1$ $x + iy - 2i \geq 1$ $x + (y - 2)i \geq 1$ $y - 2 \geq 1$ $y - 2 \geq 1$ or $-(y - 2) \geq 1$ $y \geq 3$ $y - 2 \leq -1$ $y \leq 1$		

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) Let $z = r(\cos \theta + i \sin \theta)$ for $-\pi < \theta \leq \pi$ Now $z^5 = r^5(\cos \theta + i \sin \theta)^5$ $= r^5 \text{cis } 5\theta$ but $z^5 = -1$ $z^5 = 1(\text{cis } \pi)$ $\therefore r^5 \text{cis } 5\theta = \text{cis } \pi$ $r^5 = 1$, $5\theta = \pi + 2k, k \in \mathbb{Z}$ $\theta = \frac{\pi}{5} + \frac{2\pi}{5}k$ $= \frac{\pi}{5}(1+2k)$ $-\pi < \frac{\pi}{5}(1+2k) < \pi$ $-5 < 1+2k \leq 5$ $-6 < 2k \leq 4$ $-3 < k \leq 2$ $z = \text{cis } \frac{\pi(1+2k)}{5}$ when $k=0, z = \text{cis } \frac{\pi}{5}$ $k=1, z = \text{cis } \frac{3\pi}{5}$ $k=2, z = \text{cis } \frac{5\pi}{5} = \text{cis } \pi$ $k=-1, z = \text{cis } \left(-\frac{\pi}{5}\right)$ $k=-2, z = \text{cis } \left(-\frac{3\pi}{5}\right)$ $\therefore$ solutions are: $z = \text{cis } -\frac{3\pi}{5}, \text{cis } -\frac{\pi}{5}, -1, \text{cis } \frac{\pi}{5}, \text{cis } \frac{3\pi}{5}$	1	Many students did not leave the solution as a principle argument Marks were still awarded but remember to leave roots in that form
Alternative Method Solutions are evenly spaced around a unit circle by $\frac{2\pi}{5}$, starting at $\frac{\pi}{5}$  $\therefore z = \text{cis } \frac{\pi}{5}, \text{cis } \frac{3\pi}{5}, \text{cis } \pi, \text{cis } \frac{7\pi}{5}, \text{cis } \frac{9\pi}{5}$ but $\text{cis } \frac{7\pi}{5} = \text{cis } \frac{3\pi}{5}$ and $\text{cis } \frac{9\pi}{5} = \text{cis } -\frac{\pi}{5}$ $\therefore z = \text{cis } -\frac{3\pi}{5}, \text{cis } -\frac{\pi}{5}, \text{cis } \pi, \text{cis } \frac{\pi}{5}, \text{cis } \frac{3\pi}{5}$	3mk - Correct solutions 2mk - 3 correct solutions incl. -1 or -4 " " not incl. -1 1mk - -1 as a solution	2

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) ii)		
Sum of the roots = $-\frac{b}{a}$		
ie		
$\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} + \cos -\frac{\pi}{5} + i \sin -\frac{\pi}{5} + \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} + \cos -\frac{3\pi}{5} + i \sin -\frac{3\pi}{5} = 0$	1 mk	To show the sum of all the roots and that they equal to 0.
$\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} + \cos \frac{\pi}{5} - i \sin \frac{\pi}{5} + \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} + \cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5} - 1 = 0$	1/2 mk	A few students did not include this step
$2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{5} = 1$	1/2 mk	if previous steps are included and correct.
$2(\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}) = 1$		
$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$		
NB: As it is a 'show' question, students should include all steps in their working, particularly terms that are cancelled out.		
b) Using integration by parts:		This part was very well done.
$\int u v' dx = uv - \int v u' dx$		
For $\int x e^{-2x} dx$		
Let $u = x$ $v' = e^{-2x}$	1	
$u' = 1$ $v = -\frac{1}{2} e^{-2x}$		
$\int x e^{-2x} dx = x \times (-\frac{1}{2} e^{-2x}) - \int -\frac{1}{2} e^{-2x} \times 1 dx$	1	
$= -\frac{1}{2} x e^{-2x} + \frac{1}{2} \int e^{-2x} dx$		
$= -\frac{1}{2} x e^{-2x} + \frac{1}{2} x - \frac{1}{2} e^{-2x} + C$		
$= -\frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} + C$	1	
$= \frac{-e^{-2x}}{2} (x + \frac{1}{2}) + C$		(3)

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) We assume: If p is an integer, then p^2+6 is divisible by 4.	$\frac{1}{2}$	— mark for the assumption / negation)
So let $p^2+6=4M$ (*) where M is an integer	$\frac{1}{2}$	
<u>Case 1</u> : If p is <u>even</u> then $p=2k$ where k is an integer		
From (*) LHS = $(2k)^2+6$ = $4k^2+6$ = $4k^2+4+2$ = $4(k^2+1)+2$ which is not divisible by 4 $\therefore$ a contradiction from the assumption that p^2+6 is divisible by 4.		
<u>Case 2</u> : If p is <u>odd</u> then $p=2k+1$ From (*) LHS = $(2k+1)^2+6$ = $4k^2+4k+1+6$ = $4k^2+4k+7$ = $4k^2+4k+4+3$ = $4(k^2+k+1)+3$ which is not divisible by 4 $\therefore$ a contradiction from the assumption that p^2+6 is divisible by 4.		
$\therefore$ If p is an integer, then p^2+6 is not divisible by 4.		

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) Alternative solution 1:		
From (4) $p^2 + 6 = 4M$		
$p^2 = 4M - 6$		
$p^2 = 2(2M - 3)$	$\frac{1}{2}$	
$\therefore p^2$ is divisible by 2		
$\therefore p$ is divisible by 2	$\frac{1}{2}$	
Let $p = 2k$ where $k \in \mathbb{Z}$		
From (*) LHS = $(2k)^2 + 6$		
$= 4k^2 + 6$		
$= 4k^2 + 4 + 2$		
$= 4(k^2 + 1) + 2$	1	
which is not divisible by 4		
but $p^2 + 6 = 4M$		
$\therefore$ a contradiction from the assumption.	$\frac{1}{2}$	(3)
$\therefore$ If p is an integer, $p^2 + 6$ is not divisible by 4.		
NB: Many students got to		
$p^2 = 2(2M - 3)$	$\frac{1}{2}$	
then took the square root to make p		
the subject, i.e. $p = \pm\sqrt{2(2M - 3)}$		
then aimed to prove that		
p was not an integer (or irrational)		
and ended the proof there saying		
it was a contradiction.		
This proof is <u>not</u> correct because		
the negation of the original		
statement is that: if <u>p is an integer</u>		
then $p^2 + 6$ is divisible by 4.		
It is already given that p is an		
integer.		

No marks were given for any subsequent working after this line.

13

[illegible]

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) i) $P(-1, 2, 3)$ $Q(-2, 4, 2)$ $\therefore \vec{PQ} = \vec{OQ} - \vec{OP}$ $= \begin{pmatrix} -2 \\ 4 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$		
$\therefore \vec{PQ}$ is the direction vector		
$\therefore \vec{r} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ or $\vec{r} = \begin{pmatrix} -2 \\ 4 \\ 2 \end{pmatrix} + t \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$		
ii) The angle between the direction vectors is $\cos^{-1} \frac{1}{6}$. So if the direction vectors are: $\vec{d}_1 = \begin{pmatrix} -1 \\ 6 \\ 2 \end{pmatrix} \quad \text{and} \quad \vec{d}_2 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix},$ then $\cos \theta = \frac{\begin{pmatrix} -1 \\ 6 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}}{\left \begin{pmatrix} -1 \\ 6 \\ 2 \end{pmatrix} \right \cdot \left \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \right }$ $\frac{1}{6} = \frac{1 + 26 - 2}{6}$ $1 = 1 + 26 - 2$ $26 = 2$ $b = 1$		
		$\frac{1}{2}$ mark
		$\frac{1}{2}$ mark
		①
		2 • Writing that if $\cos^{-1} \frac{1}{6} = \theta$ then $\frac{1}{6} = \cos \theta$ • Finding the dot product • Find the magnitude of $\vec{d}_1$ and $\vec{d}_2$
		For answer ③

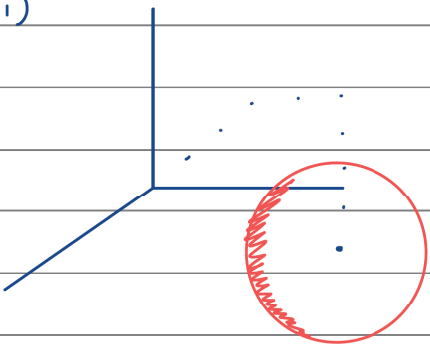
MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $\alpha + \beta = -\frac{b}{a}$ $= -\frac{-(3-i)}{4+3i}$ $= \frac{3-i}{4+3i} \times \frac{4-3i}{4-3i}$ $= \frac{12-9i-4i-3}{16+9}$ $= \frac{9-13i}{25}$ $= \frac{9}{25} - \frac{13}{25}i \quad \text{quad 4}$ $ \alpha + \beta = \sqrt{\left(\frac{9}{25}\right)^2 + \left(\frac{13}{25}\right)^2}$ $= \sqrt{\frac{250}{225}}$ $= \frac{5\sqrt{10}}{25}$ $= \frac{\sqrt{10}}{5} \quad \text{or} \quad \sqrt{\frac{2}{5}}$ $\arg(\alpha + \beta) = -\tan^{-1}\left(\frac{13/25}{9/25}\right) \quad \text{quad 4}$ $= -\tan^{-1}\left(\frac{13}{9}\right)$		

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) (i) $t = \tan \frac{\theta}{2}$ $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$ $\sin^2 \theta + \cos^2 \theta = 1$ $\therefore \tan^2 \theta + 1 = \sec^2 \theta$ $= \frac{1}{2} (\tan^2 \frac{\theta}{2} + 1)$ $= \frac{1}{2} (t^2 + 1)$ $\frac{dt}{d\theta} = \frac{t^2 + 1}{2}$ $\therefore \frac{d\theta}{dt} = \frac{2}{1+t^2}$	(1) (1)	correct derivative Correctly showing $\frac{d\theta}{dt} = \frac{2}{1+t^2}$
(ii) $\int_0^{\frac{\pi}{2}} \frac{3}{8 \cos \theta + 10} d\theta$ $t = \tan \frac{\theta}{2} \quad \theta = \frac{\pi}{2} \rightarrow t = \tan \frac{\pi}{4}$ $\frac{d\theta}{dt} = \frac{2}{1+t^2} \quad t = 1$ $\theta = 0 \rightarrow t = \tan 0$ $t = 0$ $\therefore d\theta = \frac{2}{1+t^2} dt$ $= \int_0^1 \left(\frac{3}{8 \left(\frac{1-t^2}{1+t^2} \right) + 10} \times \frac{2}{1+t^2} \right) dt$ $= 3 \int_0^1 \frac{1}{4(1-t^2) + 5(1+t^2)} dt$ $= 3 \int_0^1 \frac{1}{9+t^2} dt$ $= 3 \left[\frac{1}{3} \tan^{-1} \left(\frac{t}{3} \right) \right]_0^1$ $= \left[\tan^{-1} \left(\frac{t}{3} \right) \right]_0^1$ $= \tan^{-1} \frac{1}{3} - \tan^{-1} 0$ $= \tan^{-1} \frac{1}{3}$ $\doteq 0.322$	(1) mark (1) mark (1) mark	t-values correct integral in terms of t. Correct answer. (½ mark if not evaluated)

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c)(i)  $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ $\vec{r} = \begin{bmatrix} 3+\lambda \\ -2-\lambda \\ 1-\lambda \end{bmatrix}$ Where does the sphere intersect L $\therefore \begin{cases} x = 3+\lambda \\ y = -2-\lambda \\ z = 1-\lambda \end{cases} \quad \text{Sub IN} \quad (x-3)^2 + (y+2)^2 + (z-4)^2 = 18$ $(3+\lambda-3)^2 + (-2-\lambda+2)^2 + (1-\lambda-4)^2 = 18$ $\lambda^2 + \lambda^2 + (-3-\lambda)^2 = 18$ $2\lambda^2 + 9 + 6\lambda + \lambda^2 = 18$ $3\lambda^2 + 6\lambda - 9 = 0$ $\lambda^2 + 2\lambda - 3 = 0$ $(\lambda+3)(\lambda-1) = 0$ $\therefore \lambda = -3 \text{ or } \lambda = 1$ Let A be the point for when $\lambda = -3$ $\begin{pmatrix} 3-3 \\ -2+3 \\ 1+3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$ Let B be the point for when $\lambda = 1$ $\begin{pmatrix} 3+1 \\ -2-1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 0 \end{pmatrix}$ So the line intersects the sphere at A(0, 1, 4) and B(4, -3, 0) Centre $\left\{ \frac{0+4}{2}, \frac{1-3}{2}, \frac{4+0}{2} \right\}$ C(2, -1, 2) $d_{AB} = \sqrt{(4-0)^2 + (-3-1)^2 + (0-4)^2}$ $= \sqrt{16+16+16}$	① ① ①	Correctly using equations to solve for λ. Correctly finding A and B. Correct centre ✓ +

MATHEMATICS EXTENSION 2 – QUESTION 14

[illegible]

MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) <u>step 1</u> - Base case - Prove true for $n=1$		
LHS = $n!$		
$= 1!$		
$= 1$		
RHS = 2^{n-1}		
$= 2^{1-1}$		
$= 2^0$		
$= 1$	$\frac{1}{2}$	Correctly showing the base case
$\therefore \text{LHS} \geq \text{RHS}$		
$\therefore$ True for $n=1$		
<u>Step 2</u> - Inductive hypothesis - Assume true for $n=k$		
i.e. $k! \geq 2^{k-1}$, $k \in \mathbb{Z}^+$	$\frac{1}{2}$	Correctly states the assumption
<u>Step 3</u> - Inductive step - Prove true for $n=k+1$		
i.e. Prove $k! (k+1) \geq 2^{k+1-1}$		
$(k+1)! \geq 2^k$	$\frac{1}{2}$	Correctly stating what is to be proved.
Consider $(k+1)! - 2^k$		
$= k! (k+1) - 2^k$		
$\geq 2^{k-1} (k+1) - 2^k$		
$= 2^{k-1} (k+1 - 2)$		
$= 2^{k-1} (k-1)$		
≥ 0 since $k \in \mathbb{Z}^+$ and $2^{k-1} > 0$ for all k		
$\therefore (k+1)! - 2^k \geq 0$		
$(k+1)! \geq 2^k$		
$\therefore$ true for $n=k+1$ if true for $n=k$		
<u>Step 4</u> $\therefore$ By the principle of mathematical induction it is true for all $n \in \mathbb{Z}^+$		

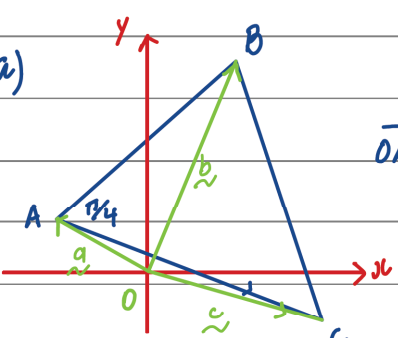
MATHEMATICS EXTENSION 2 – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(OR) <u>Step 3</u> Prove $(k+1)! \geq 2^k$ LHS = $(k+1)!$ $= k!(k+1)$ $\geq 2^{k-1}(k+1)$ $= k2^{k-1} + 2^{k-1}$ $\geq 2^{k-1} + 2^{k-1}$ $= 2 \cdot 2^{k-1}$ $= 2^k$ $\therefore (k+1)! \geq 2^k$		
$k! \geq 2^{k-1}$ $10 \geq 8$ $10 \times 2 \geq 8$ or $\geq 2^{k-1}(1+1)$ since $k \in \mathbb{Z}^+ \therefore k \geq 1$ $= 2^{k-1} \cdot 2$ $= 2^k$		

MATHEMATICS EXTENSION 2 – QUESTION 15

[illegible]

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) a)  $\vec{OA} + \vec{AC} = \vec{OC}$ $\vec{AC} = \sqrt{2} \vec{AB}$ $\vec{AB} = \underline{b} - \underline{a}$ $\vec{AC} = \underline{c} - \underline{a}$		
To get $\vec{AB}$ we rotate vector $\vec{AC}$ anticlockwise by $\frac{\pi}{4}$ and reduce it by a factor of $\frac{1}{\sqrt{2}}$ since $AC = \sqrt{2} AB$ $\therefore \vec{AB} = \frac{1}{\sqrt{2}} \vec{AC} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$ $AB = \frac{1}{\sqrt{2}} AC$ $\underline{b} - \underline{a} = \frac{1}{\sqrt{2}} (\underline{c} - \underline{a}) (\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i)$ $\underline{b} - \underline{a} = (\underline{c} - \underline{a}) (\frac{1}{2} + \frac{1}{2} i)$ $2(\underline{b} - \underline{a}) = (\underline{c} - \underline{a})(1 + i)$ $2\underline{b} - 2\underline{a} = \underline{c} + \underline{c}i - \underline{a} - \underline{a}i$ $2\underline{b} - \underline{a} + \underline{a}i = \underline{c}(1 + i)$ $\underline{c} = \frac{2\underline{b} - \underline{a} + \underline{a}i}{1 + i} \quad \times (1 - i)$ $\underline{c} = \frac{(2\underline{b} - \underline{a})(1 - i) + \underline{a}i(1 - i)}{1 - i^2}$ $\underline{c} = \frac{2\underline{b} - 2\underline{b}i - \underline{a} + \underline{a}i + \underline{a}i + \underline{a}}{2}$ $\underline{c} = \frac{\cancel{2\underline{b}}(1 - i) + \cancel{2\underline{a}}i}{2}$ $\underline{c} = \underline{b}(1 - i) + \underline{a}i$	① ① ①	Some correct progress Correct rotation and reduction Correct process to show result.

MATHEMATICS EXTENSION 2 – QUESTION		
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
<u>METHOD 2</u>		
To get $\vec{AC}$, rotate $\vec{AB}$ clockwise by $\frac{\pi}{4}$ and enlarge by a factor of $\sqrt{2}$		
$\vec{AC} = \sqrt{2} \vec{AB} (\cos(-\frac{\pi}{4}) + \sin(-\frac{\pi}{4})i)$		
$= \vec{AB} \times \sqrt{2} (\cos \frac{\pi}{4} - \sin \frac{\pi}{4}i)$		
$\vec{AC} = \vec{AB} \times \sqrt{2} (\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i)$		
$\underline{c} - \underline{a} = (\underline{b} - \underline{a}) (1 - i)$		
$\underline{c} = \underline{b} - \underline{b}i - \underline{a} + \underline{a}i + \underline{a}$		
$\underline{c} = \underline{b} (1 - i) + \underline{a}i$		
$\vec{AB} \cdot \vec{AC} = \vec{AB} \vec{AC} \cos \frac{\pi}{4}$		
$(\underline{b} - \underline{a}) \cdot (\underline{c} - \underline{a}) = AB \times AC \times (\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i)$		
$= AB \times \sqrt{2} AB (\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i)$		
$\underline{b} \cdot \underline{c} - \underline{b} \cdot \underline{a} - \underline{a} \cdot \underline{c} + \underline{a} \cdot \underline{a} = AB^2 (1 + i)$		

METHOD 2

To get $\vec{AC}$, rotate $\vec{AB}$ clockwise by $\frac{\pi}{4}$ and enlarge by a factor of $\sqrt{2}$

$$\begin{aligned}\vec{AC} &= \sqrt{2} \vec{AB} (\cos(-\frac{\pi}{4}) + \sin(-\frac{\pi}{4})i) \\ &= \vec{AB} \times \sqrt{2} (\cos \frac{\pi}{4} - \sin \frac{\pi}{4}i)\end{aligned}$$

$$\vec{AC} = \vec{AB} \times \sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i \right)$$

$$\frac{c-a}{2} = \frac{(b-a)}{2} (1-i)$$

$$c = \underbrace{b}_{\sim} - \underbrace{bi}_{\sim} - \underbrace{a}_{\sim} + \underbrace{ai}_{\sim} + \underbrace{a}_{\sim}$$

$$\underline{c} = \underline{b}(1-i) + \underline{a}i$$

$$\vec{AB} \cdot \vec{AC} = |\vec{AB}| |\vec{AC}| \cos \frac{\pi}{4}$$

$$\begin{aligned}(\underline{b-a}) \cdot (\underline{c-a}) &= AB \times AC \times \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) \\ &= AB \times \sqrt{2} AB \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)\end{aligned}$$

$$\underline{b \cdot c} - \underline{b \cdot a} - \underline{a \cdot c} + \underline{a \cdot a} = AB^2 (1+i)$$

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) $\int_2^3 \sqrt{\frac{x-2}{4-x}} dx$ $x = 2 + 2 \cos^2 \theta$ $\frac{dx}{d\theta} = -4(\cos \theta) \sin \theta$ $x = 3, \quad 3 = 2 + 2 \cos^2 \theta$ $\cos^2 \theta = \frac{1}{2}$ $\cos \theta = \frac{1}{\sqrt{2}}$ $\theta = \frac{\pi}{4}$ $x = 2, \quad 2 = 2 + 2 \cos^2 \theta$ $\cos^2 \theta = 0$ $\theta = \frac{\pi}{2}$	02 $x = 2 + 2 \times \frac{1}{2} (\cos 2\theta + 1)$ $x = 2 + \cos 2\theta + 1$ $x = 3 + \cos 2\theta$ $\frac{dx}{d\theta} = -2 \sin 2\theta$ $= -2 \times 2 \sin \theta \cos \theta$ $= -4 \sin \theta \cos \theta$ ① Correct bounds	
$= \int_{\pi/2}^{\pi/4} \frac{\sqrt{2 \cos^2 \theta}}{\sqrt{4 - 2 - 2 \cos^2 \theta}} \times -4 \cos \theta \sin \theta d\theta$ $= \int_{\pi/2}^{\pi/4} \frac{\sqrt{2 \cos^2 \theta}}{\sqrt{2(1 - \cos^2 \theta)}} \times -4 \cos \theta \sin \theta d\theta$ $= \int_{\pi/2}^{\pi/4} \frac{\sqrt{\cos^2 \theta}}{\sqrt{\sin^2 \theta}} \times -4 \cos \theta \sin \theta d\theta$	① Correct set up of integral.	
$= \int_{\pi/2}^{\pi/4} \frac{\cos \theta}{\sin \theta} \times -4 \cos \theta \sin \theta d\theta$ $= -4 \int_{\pi/2}^{\pi/4} \cos^2 \theta d\theta$ $\cos 2\theta = 2 \cos^2 \theta - 1$ $\cos^2 \theta = \frac{1}{2} (\cos 2\theta + 1)$	① Correct simplification of integral.	
$= \frac{4}{2} \int_{\pi/4}^{\pi/2} \cos 2\theta + 1 d\theta$ $= 2 \left[\frac{1}{2} \sin 2\theta + \theta \right]_{\pi/4}^{\pi/2}$ $= 2 \left[\frac{1}{2} \sin \pi + \frac{\pi}{2} - \left(\frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right) \right]$ $= 2 \left[\frac{\pi}{2} - \frac{1}{2} \times 1 - \frac{\pi}{4} \right]$ $= \pi - 1 - \frac{\pi}{2}$ $= \frac{\pi}{2} - 1$	① Correctly integrates and gives correct solution. Some half marks awarded for minor errors	

MATHEMATICS EXTENSION 2 – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
d) $I_n = \int_0^{\pi/2} x^n \cos x \, dx$ $u = x^n \quad v' = \cos x$ $u' = nx^{n-1} \quad v = \sin x$ $= \left[x^n \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} x^{n-1} \sin x \, dx$ $= \left(\frac{\pi}{2}\right)^n \sin \frac{\pi}{2} - 0 - n \left[\left[x^{n-1} \cos x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} x^{n-2} \cos x \, dx \right]$ $= \left(\frac{\pi}{2}\right)^n - n \left[\left(\frac{\pi}{2}\right)^{n-1} \cos \frac{\pi}{2} - 0 \right] + (n-1) I_{n-2}$ $= \left(\frac{\pi}{2}\right)^n - n(0-0) - n(n-1) I_{n-2}$ $I_n = \left(\frac{\pi}{2}\right)^n - n(n-1) I_{n-2} \quad n \geq 2$	① ① ①	Correct use of integration by parts applied. Correct second use of integration by parts applied. Correctly proves I_n.
(ii) $I_4 = \left(\frac{\pi}{2}\right)^4 - 4(4-1) I_2$ $= \left(\frac{\pi}{2}\right)^4 - 12 \left[\left(\frac{\pi}{2}\right)^2 - 2(2-1) I_0 \right]$ $= \left(\frac{\pi}{2}\right)^4 - 12 \left[\left(\frac{\pi}{2}\right)^2 - 2 \times \int_0^{\pi/2} x \cos x \, dx \right]$ $= \left(\frac{\pi}{2}\right)^4 - 12 \left[\left(\frac{\pi}{2}\right)^2 - 2 \times \left[\sin x \right]_0^{\pi/2} \right]$ $= \left(\frac{\pi}{2}\right)^4 - 12 \left[\left(\frac{\pi}{2}\right)^2 - 2 \times 1 \right]$ $= \left(\frac{\pi}{2}\right)^4 - 12 \left(\frac{\pi}{2}\right)^2 + 24$	① ① ①	Correct substitution of $n=4$ into I_n Convincingly proves result, including when $n=0$.

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) i) Aim: To prove $\int_0^a f(x) dx = \int_0^a f(a-x) dx$		This part was not done very well
Let $u = a - x$		
$\frac{du}{dx} = -1$		
$dx = -du$		
When $x = a$, $u = 0$ $x = 0$, $u = a$		
Now		
$RHS = \int_0^a f(a-x) dx$		
$= \int_a^0 f(u) \cdot -du$	1 mk	$\frac{1}{2}$ - correct limits of integration $\frac{1}{2}$ - correct u substitution
$= \int_0^a f(u) du$	$\frac{1}{2}$	
$= \int_0^a f(x) dx$ (since u, x are dummy variables)	$\frac{1}{2}$	
$= LHS$		(2)
Method 2		
Let $u = a - x$, then $x = a - u \Rightarrow dx = -du$ $x = 0 \Rightarrow u = a$ $x = a \Rightarrow u = 0$		
$LHS = \int_0^a f(x) dx = \int_a^0 f(a-u) (-du)$ $= - \int_a^0 f(a-u) du$ $= \int_0^a f(a-u) du$ $= \int_0^a f(a-x) dx, \text{ since } u, x \text{ are dummy variables}$ $= RHS \quad \text{as required}$		

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) ii) Let $x = \sin \theta$ $dx = \cos \theta d\theta$ $x = 0 \Rightarrow \theta = 0,$ $x = 1 \Rightarrow \theta = \frac{\pi}{2}$ Let $I_1 = \int_0^1 \frac{dx}{x + \sqrt{1-x^2}}$, then $I_1 = \int_0^{\pi/2} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta$ $= \int_0^{\pi/2} \frac{\cos(\pi/2 - \theta)}{\sin(\pi/2 - \theta) + \cos(\pi/2 - \theta)} d\theta, \text{ using part (i)}$ $= \int_0^{\pi/2} \frac{\sin \theta}{\cos \theta + \sin \theta} d\theta$ $= I_2$ Since $I_1 + I_2 = \int_0^{\pi/2} \frac{\cos \theta + \sin \theta}{\sin \theta + \cos \theta} d\theta$ $= \int_0^{\pi/2} 1 d\theta$ $= [\theta]$ $= \left[\frac{\pi}{2} - 0 \right] = \frac{\pi}{2}$ But $I_1 = I_2$ $2I_1 = \frac{\pi}{2}$ $I_1 = \frac{\pi}{4}$ $\int_0^1 \frac{dx}{x + \sqrt{1-x^2}} = \frac{\pi}{4}$	1 1 1/2 1/2	1/2 - correct limits of integration 1/2 - correct integrand after substitution 1 - correct substitution of property in (i). 3
b) $\vec{PO} = \begin{pmatrix} -8 \\ -6 \\ 0 \end{pmatrix}$ $\vec{PR} = \begin{pmatrix} x-8 \\ y-6 \\ z \end{pmatrix}$ Using the dot product, $\vec{PO} \cdot \vec{PR} = \vec{PO} \vec{PR} \cos \frac{\pi}{3} \quad \dots (*)$ $\vec{PO} \cdot \vec{PR} = -8(x-8) - 6(y-6) + 0 \times z$ $= -8x + 64 - 6y + 36$ $= -8x - 6y + 100$ and $\vec{PO} = \sqrt{(-8)^2 + (-6)^2}$ $= \sqrt{64 + 36}$ $= \sqrt{100}$ $= 10 \quad \text{and} \quad \vec{PR} = 10 \text{ (given)}$	1/2	some students wrote $\vec{PO}$ as $\begin{pmatrix} 8 \\ 6 \\ 0 \end{pmatrix}$ which brought on errors.

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
From dot product (*) $-8x - 6y + 100 = 10 \times 10 \times \cos \frac{\pi}{3}$ $-8x - 6y + 100 = 100 \times \frac{1}{2}$ $-8x - 6y + 100 = 50$ $\therefore 4x + 3y = 25 \quad \text{--- (1)}$	1/2	
Also $\vec{QO} = \begin{pmatrix} -6 \\ -8 \\ 0 \end{pmatrix}$ and $\vec{QR} = \begin{pmatrix} x-6 \\ y-8 \\ z \end{pmatrix}$		
Dot product $\vec{QO} \cdot \vec{QR} = \vec{QO} \vec{QR} \cos \frac{\pi}{3} \quad (* *)$		
Now $\vec{QO} \cdot \vec{QR} = -6(x-6) - 8(y-8)$ $= -6x + 36 - 8y + 64$ $= -6x - 8y + 100$	1/2	
and $\vec{QO} = \sqrt{(-6)^2 + (-8)^2}$ $= 10$		
and $\vec{QR} = 10$ given sub (* *)		
$-6x - 8y + 100 = 10 \times 10 \times \frac{1}{2}$ $-6x - 8y + 100 = 50$ $-6x - 8y = -50$ $3x + 4y = 25 \quad \dots \dots (2)$	1/2	
(1) $\times 4$, (2) $\times 3$ $16x + 12y = 100 \quad \dots \dots (3)$ $9x + 12y = 75 \quad \dots \dots (4)$		
(3) - (4) $7x = 25$ $x = \frac{25}{7} \quad \text{sub in (1)}$		
$4\left(\frac{25}{7}\right) + 3y = 25$ $\frac{100}{7} + 3y = 25$		

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$100 + 21y = 175$ $21y = 75$ $y = \frac{75}{21}$ $y = \frac{25}{7}$		
Now $\vec{QR} = \sqrt{(x-6)^2 + (y-8)^2 + z^2}$ sub $x=y=\frac{25}{7}$ and $\vec{QR} =10$		
$10 = \sqrt{\left(\frac{25}{7}-6\right)^2 + \left(\frac{25}{7}-8\right)^2 + z^2}$		
$10 = \sqrt{\frac{289}{49} + \frac{961}{49} + z^2}$		
$= \sqrt{\frac{1250}{49} + z^2}$		
Squaring both sides		
$100 = \frac{1250}{49} + z^2$		
$z^2 = \frac{3650}{49}$		
$z = \frac{\sqrt{3650}}{7} \quad \text{as } z > 0$		
$= \frac{5\sqrt{146}}{7}$		
$\therefore R \text{ is } \left(\frac{25}{7}, \frac{25}{7}, \frac{5\sqrt{146}}{7}\right)$		
<u>NOTE:</u> Many students identified that they needed to find $\vec{PR} + \vec{QR}$, they found their magnitudes and they equated them, finding that $x=y$. However students did not know what to do next, it was best for them to use the dot product to find x and y then use both the values to find z. See Alternative solution below		

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Alternative Solution 16 b)		
$\vec{PR} = \begin{pmatrix} x-8 \\ y-6 \\ z-0 \end{pmatrix} \quad \vec{QR} = \begin{pmatrix} x-6 \\ y-8 \\ z \end{pmatrix}$ $= \begin{pmatrix} x-8 \\ y-6 \\ z \end{pmatrix}$		
but $ \vec{PR} = \vec{QR} = 10$		
Now $ \vec{PR} = \sqrt{(x-8)^2 + (y-6)^2 + z^2}$		
$10 = \sqrt{(x-8)^2 + (y-6)^2 + z^2}$		
$100 = (x-8)^2 + (y-6)^2 + z^2 \quad \dots (1)$		
Also		
$ \vec{QR} = \sqrt{(x-6)^2 + (y-8)^2 + z^2}$		
$10 = \sqrt{(x-6)^2 + (y-8)^2 + z^2}$		
$100 = (x-6)^2 + (y-8)^2 + z^2 \quad \dots (2)$		
$(1) = (2)$		
$(x-8)^2 + (y-6)^2 + z^2 = (x-6)^2 + (y-8)^2 + z^2$		
$x^2 - 16x + 64 + y^2 - 12y + 36 = x^2 - 12x + 36 + y^2 - 16y + 64$	1	
$-16x - 12y = -12x - 16y$		
$-4x = -4y$		
$x = y$	1	
Now $\vec{QD} = \begin{pmatrix} -6 \\ -8 \\ 0 \end{pmatrix}$		

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Using the dot product $\vec{QR} \cdot \vec{QO} = \vec{QR} \cdot \vec{QO} \cos \frac{\pi}{3}$		
$-6(x-6) - 8(y-8) = 10 \times 10 \times \frac{1}{2}$ $-6x + 36 - 8y + 64 = 50$ $-6x - 8y = -50$ $6x + 8y = 50$ $3x + 4y = 25$	$\frac{1}{2}$	
but $x = y$		
ie $3x + 4x = 25$ $7x = 25$ $x = \frac{25}{7}$ and $y = \frac{25}{7}$	$\frac{1}{2}$	
For z , sub x & y in ①		
$100 = \left(\frac{25}{7} - 8\right)^2 + \left(\frac{25}{7} - 6\right)^2 + z^2$ $100 = \frac{961}{49} + \frac{289}{49} + z^2$ $z^2 = \frac{3650}{49}$ $z = \pm \sqrt{\frac{3650}{49}}$ but $z > 0$ $\therefore z = \sqrt{\frac{3650}{49}}$ $= \frac{5\sqrt{146}}{7}$		
$\therefore R$ is $\left(\frac{25}{7}, \frac{25}{7}, \frac{5\sqrt{146}}{7}\right)$	1	

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) Given $\frac{x+y}{2} \geq \sqrt{xy}$ Let $x \rightarrow a, y \rightarrow b$ $\therefore \frac{a+b}{2} \geq \sqrt{ab}$ i.e. $a+b \geq 2\sqrt{ab}$ --- (1) Similarly $a+c \geq 2\sqrt{ac}$ --- (2) $b+c \geq 2\sqrt{bc}$ --- (3) By adding (1), (2) and (3) we get $a+b+a+c+b+c \geq 2\sqrt{ab} + 2\sqrt{ac} + 2\sqrt{bc}$ $2a+2b+2c \geq 2(\sqrt{ab} + \sqrt{ac} + \sqrt{bc})$ $2(a+b+c) \geq 2(\sqrt{ab} + \sqrt{ac} + \sqrt{bc})$ $a+b+c \geq \sqrt{ab} + \sqrt{ac} + \sqrt{bc}$ --- (4)		This part was not well done
Now consider the expansion: $(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 = a+b+c + 2\sqrt{ab} + 2\sqrt{ac} + 2\sqrt{bc}$ --- (5) Add $2\sqrt{ab} + 2\sqrt{ac} + 2\sqrt{bc}$ to both sides of (4) we get $a+b+c + 2\sqrt{ab} + 2\sqrt{ac} + 2\sqrt{bc} \geq \sqrt{ab} + \sqrt{ac} + \sqrt{bc} + 2\sqrt{ab} + 2\sqrt{ac} + 2\sqrt{bc}$ $\therefore$ Substituting LHS of (5) we get $(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 \geq 3\sqrt{ab} + 3\sqrt{ac} + 3\sqrt{bc}$ $(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 \geq 3(\sqrt{ab} + \sqrt{ac} + \sqrt{bc})$		
3 Marks : Full, clear and well structured proof		
2 Marks : - Adding all AM/GM inequalities and dividing both sides by 2. - considering the expansion. $(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 = a+b+c + 2\sqrt{ab} + 2\sqrt{ac} + 2\sqrt{bc}$		
1 Mark : Adding all AM/GM inequalities & dividing both sides by 2.		

3 marks

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
ii) From (i) $(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 \geq 3(\sqrt{ab} + \sqrt{ac} + \sqrt{bc}) \quad \dots (*)$ Let $a = m^6 p^6$, $b = m^6 r^6$, $c = p^6 r^6$ sub in $(*)$ $(\sqrt{m^6 p^6} + \sqrt{m^6 r^6} + \sqrt{p^6 r^6})^2 \geq 3(\sqrt{m^6 p^6 m^6 r^6} + \sqrt{m^6 p^6 p^6 r^6} + \sqrt{m^6 p^6 r^6 r^6})$ $(m^3 p^3 + m^3 r^3 + p^3 r^3)^2 \geq 3(\sqrt{m^{12} p^6 r^6} + \sqrt{m^6 p^{12} r^6} + \sqrt{m^6 p^6 r^{12}})$ $(m^3 p^3 + m^3 r^3 + p^3 r^3)^2 \geq 3(m^6 p^3 r^3 + m^3 p^6 r^3 + m^3 p^3 r^6)$ $> m^6 p^3 r^3 + m^3 p^6 r^3 + m^3 p^3 r^6$ $(m^3 p^3 + m^3 r^3 + p^3 r^3)^2 \geq m^3 p^3 r^3 (m^3 + p^3 + r^3)$		After giving the subst. for a, b and c, many students did not include the LHS of this line in their proof and went straight to the third line. (2)
NOTE: This is a proof question and requires you to show <u>ALL</u> steps. Please attempt more proof/show type questions and practise showing all steps in your solutions.		